

ACTIVITY 5 — LEARNER WORKSHEET**DMAIC Overview, Problem Statement, Scope, and Stakeholders**

TYPE	DURATION	DMAIC PHASE	MAPS TO
Core Lab	45 min	DEFINE	A1, A2

Name	
Date	

What you are doing

Step up from PDCA to the full DMAIC roadmap. Clarify what each phase delivers, where a Yellow Belt contributes most, and prepare defensible Define-phase information for the project leader.

What you will produce: A DMAIC phase table, a refined problem statement, a stakeholder map and a benefit estimate in business terms.

Tools and techniques: DMAIC roadmap, project charter, stakeholder analysis, CTQ linkage

The scenario

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder

templates/ — the worksheets you complete:

- dmaic-phase-table.csv (phase, purpose, key_deliverable, yellow_belt_support_role)
- stakeholder-map.csv (stakeholder, interest, influence_H_M_L, interest_H_M_L, engagement_approach)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

Your steps

1. Summarise all five DMAIC phases: purpose, key deliverable and Yellow Belt support role.

2. Refine the problem statement so it is specific, measurable and solution-free.
3. Identify stakeholders and classify each by influence and interest.
4. Define the project scope and state the expected benefit in business terms.
5. Link the problem back to the CTQ requirements captured in Lab 2.
6. Confirm which DMAIC phases a Yellow Belt can support most strongly.

Your working space

Complete every field. You may keep this worksheet and refer to it during the open-book assessment.

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Step 3 — Identify stakeholders and classify each by influence and interest.

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Step 5 — Link the problem back to the CTQ requirements captured in Lab 2.

Step 6 — Confirm which DMAIC phases a Yellow Belt can support most strongly.

Answers to the discussion prompts

1. Which DMAIC phase does a Yellow Belt support most strongly, and why?

2. Which stakeholder could stop this project, and how will you engage them?

3. How does the problem connect back to the CTQs you captured in Lab 2?

Reflect & discuss

Who are the high-power, low-interest stakeholders in your own work?

What business benefit would make your manager fund an improvement?

Check your work

Your problem statement passes the 'no solution named' test and every stakeholder has a defined engagement approach.